

Session 1 Practical

A. Relations

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

You will be given two numbers X and Y and you will have to define the relation between them. It can be one of the following relations :

- X greater than Y.

- X smaller than Y.

- X equals Y.

Input

The input consists of two numbers X and Y ($-10^{12} < x, y < 10^{12}$).

Output

Print one line containing the right relational operator ">", "<", "=" without quotes.

Sample test(s)

input
1 2
output
<

input
10 10
output
=

B. Lucky mom

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Our mother has N chocolates and she wants to divide them between her two children in a fair way.

So if the number of chocolates is odd she will have to take one from them.

Help us know if our mother is lucky enough to take a chocolate or not.

Input

The input consists of one integer N , ($0 < N < 2000$), which indicates the number of chocolates our mother has.

Output

Print the word "lucky" if our mother is lucky enough to take a chocolate otherwise print "unlucky" without the quotes.

Sample test(s)

input
5
output
lucky

input
4
output
unlucky

C. Calculator

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Koko is a new student in FCIS and he has attended the first session in acmASCIS level 1. He wants to make a simple calculator. He wants his calculator to be able to add, subtract, multiply and divide two integers.

Input

The input consists of an integer X then symbol('+', '-', '*' or '/') without the quotes) then another integer Y. ($0 < |X|, |Y| < 10^4$).

Output

For each test case, print the result of the calculation made by koko's calculator.

Sample test(s)

input
3+5
output
8

input
5*5
output
25

input
25/3
output
8

Note

Note that the division in this calculator is an integer division.

D. Rectangle

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output



Given two coordinates in a rectangle "ABCD", you are asked to calculate the other two coordinates.

Input

The first line contains four space-separated integers x_1, y_1, x_2, y_2 ($-100 \leq x_1, y_1, x_2, y_2 \leq 100$), ($x_1 \neq x_2$ and $y_1 \neq y_2$), where (x_1, y_1) and (x_2, y_2) are point D,B or B,D.

Output

Print the four coordinates in this format " $(A_x, A_y)(B_x, B_y)(C_x, C_y)(D_x, D_y)$ " without the quotes.

Sample test(s)

input
0 0 10 5
output
(0,5)(10,5)(10,0)(0,0)

input
10 5 0 0
output
(0,5)(10,5)(10,0)(0,0)

Note

Point A is top left and Line AB is parallel to x axis.

E. Sprint

time limit per test: 1 second

memory limit per test: 64 megabytes

input: standard input

output: standard output

Ahmed is so competitive, in this weekend he challenged his friends Ali and Mohamed to reach the playground before them.

And because of his fitness, he won the race.

If you have the time that everyone took to reach the playground, can you guess which one belongs to Ahmed?!

Input

Three integers A, B and C ($1 \leq A, B, C \leq 10000$) representing the time that everyone took to reach the playground in seconds.

Output

Print the time Ahmed took to reach the playground.

Sample test(s)

input
100 140 130
output
100

input
233 198 201
output
198